Apply to be a Science Corps Fellow - Entry #3357

Name

Tuhin Bandopadhyay

Email

tuhinb2@illinois.edu

Gender

Male

Residence: City (State), Country

Champaign, IL, USA

Nationality

Indian

When did you or do you expect to complete your PhD?

January 2026

Please tell us briefly about your university education (institution, city, country, year)

I am a doctoral candidate in the Department of Aerospace Engineering at the University of Illinois at Urbana-Champaign (UIUC), joined in August 2020 and anticipate graduating in Early 2026. I did dual degree (B.Tech+ M.Tech) from IIT Kharagpur in India, from 2015 to 2020.

Why would you like to be a Science Corps Fellow?

I would like to be a Science Corps Fellow because I am passionate about using my scientific training to make meaningful contributions to global education. As a soon-to-be PhD graduate in experimental fluid mechanics, I've spent years studying particle-laden turbulent flows and developing robust tools for experimental data analysis. While this work has advanced my technical expertise, what has consistently brought me the greatest fulfillment is mentoring students, simplifying complex ideas, and engaging with communities through science.

Throughout my PhD, I've led lab sessions, mentored undergraduate researchers, and given outreach presentations aimed at making fluid dynamics accessible to broader audiences. These experiences taught me that effective science education is not just about delivering content—it's about inspiring curiosity and building confidence. I see the Science Corps Fellowship as a chance to apply these skills where they can have long-term impact: in classrooms where educational resources may be limited, but student potential is limitless.

The opportunity to work at the Central Visayan Institute Foundation and collaborate with local teachers to develop and implement science curriculum is particularly exciting. I am eager to co-create engaging, inquiry-based lessons that connect scientific principles to students' everyday lives. My background in experimental work allows me to design low-cost, hands-on demonstrations that illustrate core concepts, fostering both understanding and enthusiasm for science.

I am also drawn to the reciprocal nature of the Fellowship. Immersing myself in a new culture, learning from local educators, and adapting to a different educational system will challenge and enrich my own perspective as a scientist and teacher. I believe that science should be shaped by diverse experiences and local contexts, and this program offers a rare opportunity to grow through service and cross-cultural exchange.

Ultimately, I hope to contribute meaningfully to CVIF’s educational mission while deepening my commitment to inclusive, globally engaged science. Whether I pursue a future in academia, education, or development, this Fellowship represents the kind of impactful, human-centered work I aim to carry forward. I am confident that the skills, perspectives, and connections gained through this experience will guide me toward becoming a more thoughtful educator and scientist.

In addition to your studies, please describe any prior experience that may help you as a Science Corps Fellow.

Since my undergraduate days at IIT Kharagpur, I have remained focused on my ambition, dedicating myself to various research projects. I founded a departmental research group, "Airavat", to develop an electric flying vehicle for rescue operations and medical emergency transport. Our team secured a 1 million Rupee grant through an Institute challenge to fund the project. My efforts paid off beyond expectations when I received the ‘Best Project’ award for my thesis work at IIT Kharagpur.
To gain further exposure to research and academia in a different environment that would expand my perspectives, I embarked on doctoral studies at UIUC. I am pursuing my research in particle-laden turbulent flows, with a focus on understanding the dynamics of inertial particles that preferentially concentrate under the influence of multiscale turbulent eddies. The goal is to develop statistical and empirical relationships between the lifetime, size, and concentration of macroscopic particle clusters, ultimately creating reduced-order models for cluster evolution and establishing connections between the macroscopic dynamics of clusters and the kinematics of constituent particles. This requires extensive experimental analysis for varying particle and flow parameters, necessitating the development of USA's largest vertical particle-laden turbulent channel flow facility. I have worked on the design and development of the facility and subsequent flow characterization, performed time-resolved measurements, and developed novel data analysis tools to extract and model particle cluster evolution for varying parameters.
Working on these research projects provided me the opportunity to lead a team, supervise undergraduate students, disseminate research findings through conferences and publications, and collaborate with peers from different backgrounds; all of which I thoroughly enjoy. Driven by the urge to teach, I also grabbed the opportunity to work as a teaching assistant for the undergraduate course ‘Incompressible Flow’ for three consecutive semesters. As a TA, I taught course modules, prepared homework and corresponding solutions, developed project materials, and created demonstration videos for the students. These experiences have expanded my horizons and prepared me to become a proficient researcher and successful team leader.
I was deeply honored to receive the Graduate College Block Grant Fellowship and the Faculty Endowed Graduate Award in recognition of my academic, research, and teaching accomplishments at UIUC. I was also selected for the Mavis Future Faculty Program, a prestigious initiative that offers comprehensive training in research, teaching, and mentoring.

Where do you imagine yourself professionally in the future? Check all that apply.

Scientist in academia
In science education

Please expand on your selection above and describe where you imagine yourself professionally in the near (~2 year) and long-term (~10 year) future.

In the near future (2 years), I see myself as a postdoctoral researcher working in a collaborative, interdisciplinary academic environment, deepening my expertise in experimental fluid dynamics while expanding into adjacent areas such as environmental flows, biomedical applications, and complex multiphase systems. I aim to engage in projects that integrate experimental, computational, and data-driven approaches to tackle real-world fluid problems. Alongside research, I hope to remain actively involved in science education—whether through mentoring undergraduate and graduate students, contributing to curriculum development, or leading outreach programs that broaden participation in STEM.

During this postdoctoral period, I plan to build a strong foundation for my independent academic career. I will focus on publishing impactful research, developing my teaching philosophy, and honing my leadership and grant-writing skills. I also intend to continue exploring inclusive pedagogical practices and international collaboration models—ideals that programs like Science Corps strongly promote and which I believe are essential for building equitable scientific communities.

In the long term (10 years), I envision leading my own research group as a faculty member at a university, driving innovation in the field of fluid dynamics through multidisciplinary and application-oriented research. I aim to establish a lab that bridges traditional fluid mechanics with emerging domains like bio-inspired design, sustainable systems, and health-related flows. My goal is not only to advance knowledge but to cultivate a lab culture that values collaboration, creativity, and inclusion—one that trains students to think across disciplines and engage with global challenges.

Equally important to me is contributing to science education at both the undergraduate and pre-university levels. I see myself developing experiential learning modules, integrating real-world data into classrooms, and building partnerships with schools to create hands-on science opportunities. Programs like Science Corps affirm my belief that education and research are not separate tracks, but deeply intertwined endeavors. In my future academic role, I hope to formalize this integration by launching initiatives that connect university research with community education efforts—both locally and internationally.

Ultimately, my professional trajectory is shaped by a dual commitment: advancing cutting-edge research in fluid dynamics and ensuring that science remains accessible, inspiring, and empowering to the next generation. I view academic science not only as a platform for discovery but as a vehicle for societal impact, and I am excited to grow into a leadership role that reflects those values.

Please upload your CV (.pdf)

[CV_tuhin.pdf]

How did you find out about Science Corps?

An email list in your institution

Science Corps